

Aidan Tjon

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EDUCATION

University of California San Diego

Class of 2029

Code Hobbits, Fremont

January 2023 - June 2025

- After School Enrichment Program, Computer Science & Robotics
- Gained expertise in Python programming and robotics through hands-on projects, including plant care automation using Python.
- Explored computer security and ethical hacking fundamentals, fostering a deeper understanding of cybersecurity principles.
- Developed machine learning models using Python and TensorFlow, focusing on AI engineering applications.

PUBLICATIONS

"Algorithmic Implicit Bias"

December 2025

Cognitive NeuroEconomics @ UCSD (Medium Publication)

- Analyzed how training data bias propagates through ML systems, examining case studies including smartphone computational photography and the COMPAS recidivism algorithm.
- Synthesized research from NeurIPS proceedings on noisy label learning with Kahneman's cognitive science framework to explain systematic bias in AI systems.
- <https://medium.com/cognitive-neuroeconomics/algorithmic-implicit-bias-24ab457f8f81>

EXPERIENCE

Student Cluster Competition 25, UCSD Home Team

June 2025 - Nov 2025

- Gained hands-on experience running CPU based MLPerf Inference (v4.1) benchmark for Llama 2-70B on the SDSC Expanse HPC cluster including environment setup and debugging errors
- Applied per-tensor FP8 (e4m3) and MXFP4 quantization techniques to optimize model performance, learning how static vs. dynamic quantization impacts inference throughput and numerical stability

Youth Volunteer, Lawrence Hall of Science

April 2024 - June 2025

- Engaged visitors with interactive science exhibits, explaining complex scientific concepts in an accessible manner to all ages, providing assistance to student staff in facilitating activities at each exhibit.
- Assisted with the setup and breakdown of exhibits and event spaces, ensuring that all materials were prepared and organized efficiently.

Project Lead, Blubot

June 2023 - Present

- Raised over \$2500 through GoFundMe, self-funding the whole project.
- Led the Blu team to construct a raised bed and engineer an autonomous robot whose goal is for anyone, anywhere to grow their own food.
- <https://youtu.be/HkEAPe8vuZg>

Booth Presenter, Maker Fair

September 2023 - October 2023

- Interacted with attendees, explained electronic circuits to all ages, and demonstrated flexibility in explaining concepts, enhancing communication and presentation skills.
- <https://makerfaire.com/yearbook/projects/giant-interactive-breadboard-2023/>

SKILLS

Programming: Python, TensorFlow, Machine Learning, Deep Neural Networks

HPC & Systems: MLPerf Benchmarking, SDSC Expanse Cluster, Model Quantization (FP8, MXFP4)

Robotics & Hardware: Autonomous Systems, Plant Care Automation, Electronic Circuits

AI Ethics & Research: Algorithmic Bias Analysis, Data Quality Assessment, Technical Writing

Leadership & Fundraising: Crowdfunding Campaign Management, Project Pitching, Team Coordination, Science Communication